GAO ERIC TIANJIAN

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SUMMARY

Energy and Power Engineering undergraduate with a Computer Science minor and hands-on robotics R&D experience in SLAM, ROS 2/DDS, LiDAR, NVIDIA Jetson Orin, Linux, and robot deployment. Experienced in system integration, network/debugging, and industrial automation, with a strong interest in machine learning and edge AI deployment for robotics and intelligent systems.

EDUCATION

Beijing University of Technology — Beijing, China
B.Eng. in Energy and Power Engineering (New Energy); Minor in Computer Science — Expected Jul 2027

TECHNICAL SKILLS

Programming: Python, C; scripting, data processing, debugging
Robotics & Systems: Linux/WSL, ROS 2, DDS/CycloneDDS, Unitree SDK2, NVIDIA Jetson Orin; LiDAR, odometry, point-cloud and network troubleshooting
Industrial & Hardware: DCS/PLC basic testing and integration; multimeter, oscilloscope, vibration test bench; Factory Acceptance Testing (FAT)
Languages: Mandarin Chinese (native); English (CET-6; technical reading and day-to-day work communication)

EXPERIENCE

Unitree Robotics | Beijing, China — Sep 2026 - Present
SLAM R&D Intern
- Support deployment and integration of a robotic customer-guidance system in a bank environment, collaborating with R&D engineers on system debugging and issue isolation.
- Monitor SLAM and LiDAR nodes on NVIDIA Jetson Orin, inspect odometry and point-cloud topics, and participate in mapping and map-save interface troubleshooting.
- Configure robot/workstation networking and diagnose ROS 2/DDS communication problems; restored topic connectivity by switching to CycloneDDS.

Unitree Robotics | Hangzhou, China — Jul 2026 - Sep 2026
Technical Support Intern
- Supported SDK-based application development and troubleshooting for A2 quadruped robots using Linux, ROS 2, and Unitree SDK2.
- Diagnosed network and LiDAR issues using device status, network conditions, and logs; documented repeatable troubleshooting procedures and development notes.
- Analyzed robot deployment requirements for industrial inspection use cases across power, petrochemical, and mining environments.

SUPCON Technology — Jul 2025 - Aug 2025
Industrial Automation Intern - Nuclear Power Business Unit
- Performed basic DCS/PLC testing and system integration at the manufacturing site; recorded results under standard operating procedures and supported issue verification.
- Reviewed engineering drawings against physical assemblies and documented discrepancies for production and test teams.

Bohua Technology — Jul 2024 - Feb 2025
Production & Test / International Market Intern
- Performed board inspection, multimeter checks, oscilloscope and vibration-table testing, software configuration tests, and FAT support for industrial monitoring products.
- Supported overseas-customer FAT and researched wireless vibration sensors and predictive-maintenance solutions, connecting product capabilities with field deployment needs.

SELECTED PROJECTS & RESEARCH

Industrial Robot Predictive Maintenance Study - Mapped failure modes across reducers, bearings, motors, and encoders; organized vibration, temperature, current, and controller-log data sources and analyzed the workflow from anomaly detection to maintenance execution.
Vanadium Redox Flow Battery State-of-Health Research - Conducted literature review and research analysis on a mechanism-informed particle-swarm-optimization approach for improving an extreme learning machine (ELM) used in battery health estimation.